

Code-Layout, Readability and Reusability

Date	30 September 2026
Team ID	6ab39579211fa4fd5d28901b
Project Name	EduGenie: Google Gemini Powered Learning Assistant
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the EduGenie codebase in terms of structure, readability, modularity, error handling, and reuse. The project is organized into separate Python feature modules with a FastAPI entry point, shared model configuration, templates, static assets, and documentation.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout the code.	Yes	Python modules follow consistent indentation and formatting.
2	Proper File Structure	Files and folders are logically organized.	Yes	Separate modules, templates, static files, docs, and configuration are maintained.
3	Meaningful Variable Names	Variables reflect their purpose clearly.	Yes	Names such as question, concept, summary_length, duration_weeks and learning_goal are descriptive.
4	Function / Method Names	Functions are descriptively named.	Yes	Feature functions such as answer_question, explain_concept and generate_quiz clearly describe their purpose.
5	Code Comments	Inline and block comments explain logic where useful.	Yes	Module documentation and clear function structure support code understanding.
6	Modular Design	Code is split into reusable functions/modules.	Yes	Q&A, explanation, quiz, summary and learning-path logic are separated into modules.
7	No Redundant Code	Duplicate or unused code is removed.	Yes	Shared model settings are centralized in model_config.py and feature logic remains separated.
8	Error Handling	Exceptions and errors are handled gracefully.	Yes	Input validation and FastAPI HTTP error responses are implemented for service failures.

Reusable Components / Modules:

Code-Layout, Readability and Reusability — Continued

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level
1	main.py	Python / FastAPI	Application entry point and shared API routing for all EduGenie features.	High
2	qna.py	Python / Gemini API	Reusable Q&A; service called by the /qa endpoint.	High
3	explanation_module.py	Python / Transformers	Reusable local concept-explanation service called by the /explain endpoint.	High
4	quiz_module.py	Python / Gemini API	Reusable quiz-generation service called by the /quiz endpoint.	High
5	summary_module.py / learning_path.py	Python / Gemini API	Reusable summarization and personalized-learning services exposed through their API endpoints.	High

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	The project follows a clear modular structure with separate feature modules, templates, static assets, and documentation.
Readability	5	Feature and request names are descriptive and the code is organized by responsibility.
Reusability	5	Feature functions and model configuration are separated so services can be reused or extended.
Documentation / Comments	5	The repository contains README and phase documentation; modules also use explanatory documentation where appropriate.
Overall Score	5/5	Good overall maintainability for the current EduGenie MVP, with scope for further refinement as features grow.